

Introduction

National Hockey League (NHL) teams enter each new season with two goals: 1) to make the playoffs, and 2) to advance as far as possible through those playoffs, all in a bid to win the coveted Stanley Cup. This desire to win in the playoffs doesn't come just from an emotional standpoint, teams also have a great financial incentive to succeed. It has been reported that teams can make over \$5 million for every single playoff home game they play simply from tickets, concessions, and merchandise^{[3], [4]}. That doesn't even include less measurable effects such as additional fan interest in the team following a good playoff run. The windfall doesn't only land on the teams, however. Zac Delaney of the Edmonton Journal reports that the city of Edmonton made over \$250 million CAD (over \$180 million USD) from each of their Oilers' two previous cup runs^[5], despite neither run resulting in a Stanley Cup victory. Considering the NHL salary cap (the maximum amount of money a team is allowed to spend on all of their players in a season) is less than \$100 million, the financial benefit of having a strong playoff team becomes immediately apparent.

One key factor to making the playoffs is, naturally, how much a team spends on their players. A simple linear regression model predicting the number of points a team will get at the end of a season (the standings are determined by a team's points) using a team's total salary shows that the team's total salary has a significant effect on a team's regular season success, predicting that a team will earn an extra 1.23 points for every additional million dollars they spend on their roster. While important, money spent is far from the only predictor for who makes the playoffs. A simple logistic regression model (using money spent as the predictor) was only able to successfully predict 78% of the teams who made the playoffs over the last four years. While this seems strong, that's still over three teams a year who miss the playoffs despite spending enough money to do so. The situation gets even more dire when it comes to how well a team does in the playoffs, as total money spent cannot significantly predict how many wins a playoff team will achieve in a run.

It appears that how much money a team spends does not tell the whole story, but that how a team spends their money could also have a great impact towards being able to collect that additional playoff revenue. This project will use team salary cap data from the last four full NHL seasons to analyze ways a team can best spend their money, from what positions to spend on to how much a team should allocate towards their top end players, all in an effort to maximize the amount of playoff revenue a team is likely to make.

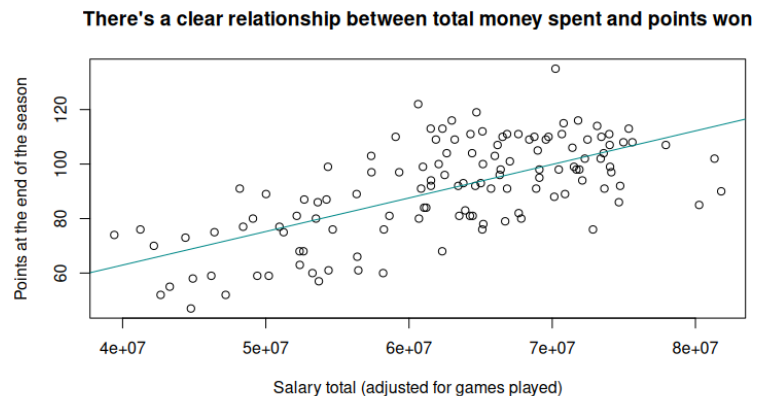


Figure 1: Relationship between total spending and season points

Business Intelligence Methods

The Business Process:

The key sections of this project roadmap are as follows:

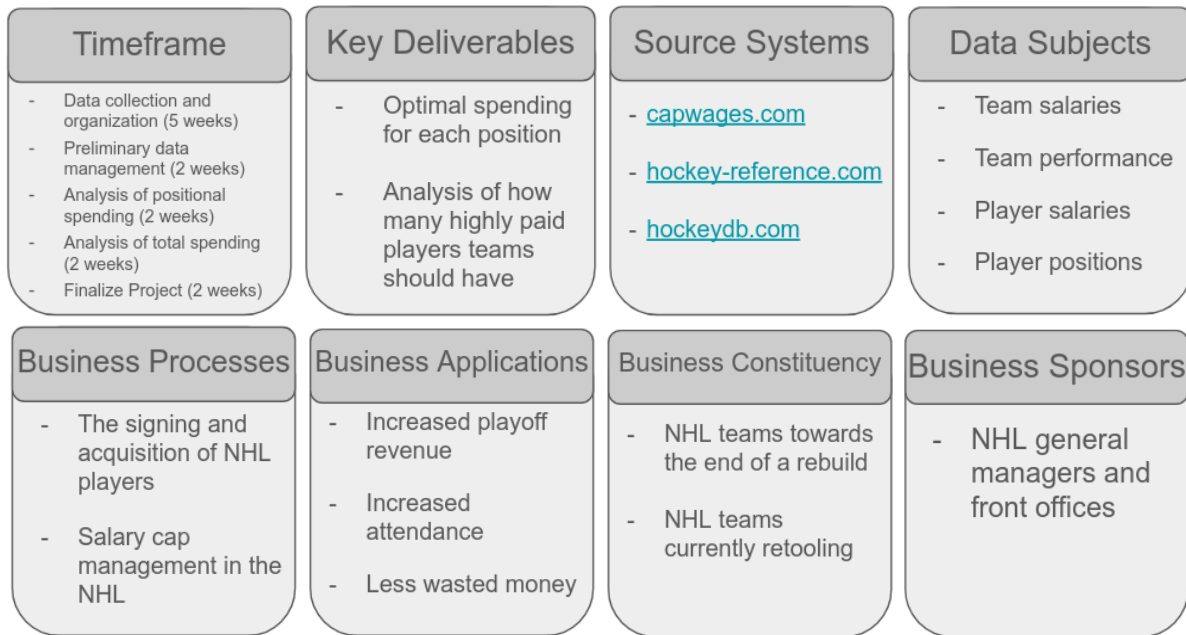


Figure 2: Project Roadmap

Some errors were found in the data acquired for the project. In particular, some of the player salaries were missing. Luckily, there were less than ten instances of this across the whole dataset, so each instance was able to be handled individually, with the actual player salary having been looked up via other sources^{[4], [5]}. After this was completed, the data was determined to be clean, consistent, and comprehensive enough for this project. The project plan was reviewed by several other students who all determined the project to be feasible. Any gaps in expertise or experience needed for this project were filled using course material or by using techniques in the famous ISLR2^[6]. This project had no financial or resource requirements beyond what was already available.

Anticipated outcomes for this project are as follows:

1. An analysis of which positions are best to focus on to maximize playoff probability
2. The optimal percentage of a team's cap that should be spent on a team's top 1/3/5/10 players.
3. A determination of whether it is worth it to have a backup goaltender who is paid more than league minimum.

Data and Data Structures:

Two primary datasets were used for this project. The first dataset was a player level dataset with the salaries of all players from the past four seasons. This dataset was acquired by first scraping all of the team rosters from hockey-reference.com^[7], and then for each player their salary for that given season was scraped from capwages.com^[8]. There were some issues during this process, particularly concerning naming conventions. The player names given from

hockey-reference were not always the exact same as the name in the URL of the corresponding player page on capwages. Most of these cases were solved using a function that removed all diacritics from a player's name, but there were still a decent number of cases where one site would use an alternative spelling, nickname, or shortened name for a player (such as using "Matt" instead of "Matthew"). These cases had to all be solved manually, which was accomplished using a piece of code that outputted the name as formatted on hockey-reference, taking the player name as formatted on capwages as input. There were also a couple players who had the exact same first and last name, which would result in both players gaining the data of one of them. These cases were nearly impossible to spot just by combing through the data, but luckily each of these cases were popular enough that they could be identified via external research. The data for each of the errant players were then looked up and manually inserted. The last issue that arose was that there were some players who did not have a capwages page. After research, it was determined that all of them did not play in the league for very long and did not sign any new contracts, so it was assumed that they each were paid the league minimum salary.

Upon investigation of this player data, it appears that contracts become less common the higher they are, with almost half of the total contracts being less than \$1 million. Since teams don't have very many high contracts, it's easy to see why it is important that teams give out those contracts to the right players.

Cap hit count by year														
By millions of dollars														
Year	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14
2021-22	606	136	95	76	60	80	44	23	18	15	7	4	1	0
2022-23	546	136	96	76	70	76	47	25	21	21	7	4	1	0
2023-24	508	125	94	80	71	74	40	29	27	22	7	5	2	0
2024-25	508	138	97	97	79	65	39	40	35	22	6	9	2	1

Table 1: Contract count by size and year

The other dataset used was a team level dataset acquired from hockey-reference.com. This dataset has information on every team over the last four seasons including how many points they got, whether they made the playoffs, and how many playoff games they won. Since all of this data was contained on the same site, the collection went much smoother..

Using the team and year columns in the player dataset, additional variables can be created for the team dataset. For example, the total cap hit for each team can be obtained by summing up the total cap hits of every player on that team that year. Simply summing up the values does lead to a problem. The issue is that teams almost never keep all of their players throughout the entire year, as players get called up, sent down, get injured, or get traded. As you can see on the right, this leads to plenty of outliers, as well as many values that are above even the highest salary cap

over this time period of \$88,000,000. To control for this, every player's cap hit was adjusted by multiplying their total hit by their games played and dividing by 82 (the total number of games played in a season) before summing them all up. Doing this removes all of the outliers and has all teams below the salary cap, as one would expect. It is worth noting that there are a number of teams below the salary floor when using the adjusted total, but that can largely be attributed to the fact that the salaries of injured players count towards the salary floor but not the salary cap, so the salaries of players who are actually playing games may not reach the floor. This adjusted total will be used in place of the regular total for team analysis.

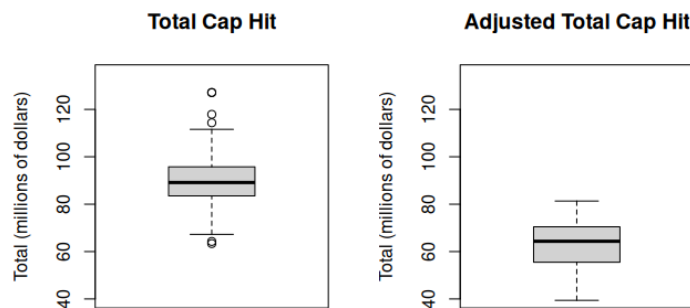


Figure 3: Boxplots of total spending before and after adjustment

Statistical Methodology

This project has two primary goals: to determine what salary distribution predictors impact a team's likelihood to make the playoffs, and then what salary distribution predictors impact the number of games a playoff team is expected to win in the playoffs. The first problem will be modeled using logistic regression, and the second will be modeled using linear regression. The reason for these decisions is that regular season success can be boiled down to whether a team made the playoffs. While teams with higher seeds often get more favorable matchups in the playoffs, the difference between the top and bottom team in the playoffs is often considered to not be as great as the difference between the bottom team in the playoffs and the top team not in the playoffs. Once the playoffs have started, however, each additional game a team plays brings in additional revenue for the team and city, and you can extend the length of your postseason by winning games.

The following predictors will be analyzed:

1. The team's total (adjusted) spending for each position
2. The team's spending on their top three, six, and nine forwards
 - a. These specific numbers were chosen because there are three forwards per line, so this is effectively analyzing how much a team is spending on their top three lines.
3. The team's spending on their top two and four defensemen
 - a. Similar to above, but this time there are only two defensemen per line
4. The amount a team spends on their backup goaltender
5. The number of players a team has that make over 10% of their total cap.
 - a. 10% here is used as a threshold for a "highly paid player"

These predictors can be used individually to determine the probability of making the playoffs (or the expected number of playoff wins) given a condition. A significance level of 0.05 will be used for all hypothesis tests.

Results

Hypothesis testing was conducted to determine if spending an above average amount on each position brought on a significant change in the expected playoff probability. The hypotheses are as follows:

$$H_0: P(\text{Pl} \mid \text{Pos} > \mu_{\text{Pos}}) = P(\text{Pl} \mid \text{Pos} \leq \mu_{\text{Pos}})$$

$$H_a: P(\text{Pl} \mid \text{Pos} > \mu_{\text{Pos}}) \neq P(\text{Pl} \mid \text{Pos} \leq \mu_{\text{Pos}}),$$

Where:

- Pl = Playoffs
- Pos = The percentage of a team's total spending spent on the position. (the test will be conducted for each position)
- μ_{Pos} = The average percentage spent on the position.

The population mean μ_0 was 0.5 for all tests. Here are the results from the tests:

Position	Sample Mean	Sample SD	Z-score	P-value
Forwards	0.531	0.503	0.500	0.617
Defensemen	0.508	0.504	0.128	0.898
Goaltenders	0.396	0.494	-1.511	0.131

Table 2: Positional hypothesis testing results

Hypothesis testing was also used to determine if the number of contracts over 10% of the cap affects a team's playoff chances. Teams had anywhere from 0 to 5 of these contracts, but since only six total teams had more than three, the groups were arranged as 0, 1, 2, and 3+. Hypotheses were stated as follows:

$$H_0: P(\text{Pl} \mid \text{Ct} = n) = \text{Pl}_0$$

$$H_a: P(\text{Pl} \mid \text{Ct} = n) \neq \text{Pl}_0,$$

Where:

- Pl = Playoffs
- Ct = Number of contracts over 10% of the cap
- n = Test group (0, 1, 2, or 3+)
- Pl_0 = The base chance of making the playoffs (50%)

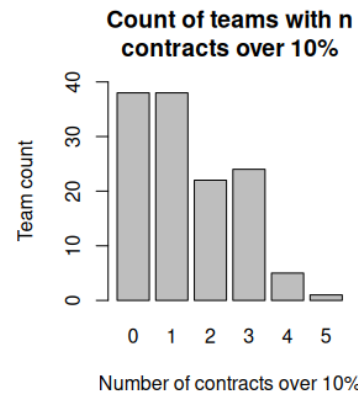


Figure 4: Number of teams with each count of contracts over 10% the salary cap

Here are the results from the tests, with significant p-values marked with an asterisk:

Contract Count	Sample Mean	Sample SD	Z-score	P-value
No Contracts	0.289	0.460	-2.596	0.009*
One Contract	0.368	0.489	-1.622	0.105
Two Contracts	0.591	0.503	0.853	0.394
Three+ Contracts	0.867	0.346	4.017	< 0.001*

Table 3: Results of hypothesis testing for counts of large contracts

Several logistic regression models with playoffs (Pl) as the response variable were trained using various combinations of the following predictor variables:

- F.perc/D.perc: The percentage of a team's total adjusted cap hit spent on forwards/defensemen.
- Tot.adj: The adjusted total cap spent by a team
- L1/L2/L3: The amount of money spent on the first/second/third line of forwards, where the first line was defined as the three highest paid forwards, the second line as the next three highest paid forwards, and similar for the third line.
- P1/P2: The amount of money spent on the first/second pairing of defensemen, where the first pairing was defined as the two highest paid defensemen, and the second pairing was defined as the next two highest paid defensemen.
- G1/G2: The primary and backup goaltender. These were determined as the two goaltenders with the most games played, respectively.

The following models were trained using these predictors, and then tested by their accuracy, sensitivity, and specificity. Model 1 was used as a base model by which all other models were compared. All models were trained using 10-fold cross validation, and the predictors were tested for individual significance using a two-tailed test. The sample size of the teams dataset was 128, so the CLT could be used for the predictors.

- Model 1: $Pl \sim Tot.adj$
- Model 2: $Pl \sim F.perc + Tot.adj$
- Model 3: $Pl \sim F.perc + D.perc + F.perc * D.perc + Tot.adj$
- Model 4: $Pl \sim L1$
- Model 5: $Pl \sim L1 + P1 + G1$
- Model 6: $Pl \sim L1 + L2 + L3 + P1 + P2 + G1 + G2$

Here are the results of each model:

Results	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Accuracy	73.25%	72.51%	75.78%	66.95%	67.25%	69.81%
Sensitivity	78.13%	76.56%	78.13%	54.06%	67.18%	68.75%
Specificity	68.75%	68.75%	73.44%	70.31%	67.18%	70.31%

Table 4: Logistic regression model results

Model 3 performed the best overall, but Model 4 was also quite strong for only having a single predictor.

Below are the results of each of the individual tests for significance:

Test	Tot.adj	F.perc	D.perc	F.perc:D.perc	L1	L2	L3	P1	P2	G1	G2
M1	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
M2	Yes	No	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
M3	Yes	No	No	No	N/A	N/A	N/A	N/A	N/A	N/A	N/A
M4	N/A	N/A	N/A	N/A	Yes	N/A	N/A	N/A	N/A	N/A	N/A
M5	N/A	N/A	N/A	N/A	Yes	N/A	N/A	No	N/A	No	N/A
M6	N/A	N/A	N/A	N/A	Yes	No	No	No	No	No	No

Table 5: Results of the tests of individual significance (at a 5% significance level)

Tot.adj and L1 were the only predictors to show significance when predicting whether a team made the playoffs. Both of these predictors showed a positive relationship with the log-odds of making the playoffs in all models where the predictors were present.

Models using the same predictors were also trained using 10-fold cross validation on the dataset of teams who made the playoffs to try and predict how many playoff wins they would achieve. These models were tested by joint significance at a 5% significance level. The results are as follows:

Results	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
S_e	4.529	4.560	4.564	4.492	4.483	4.571
R^2	0.0094	0.0122	0.0428	0.0258	0.0610	0.0887
Adjusted R^2	-0.0066	-0.0201	-0.02206	0.0101	0.0140	-0.0252
P-value	0.4464	0.6878	0.6222	0.2045	0.2834	0.6075

Table 6: Linear regression model results

Analysis of the effects of having a well paid (above the median for the playoffs) backup goaltender showed that the expected number of playoff wins if the team's backup was well paid was 4.906, while the expected number of playoff wins if the back up wasn't well paid was 6.063, a slight increase. Analysis of the distribution between the two scenarios also showed that the number of playoff wins achieved tended slightly towards the extremes when the backup goaltender was paid less than the median amount (see figure 5).

A similar analysis of the expected number of playoff wins given the number of contracts over 10% of the cap that a team gave the following results:

Playoff Wins	Nobody over 10%	One over 10%	Two over 10%	Three+ over 10%
Expectation	4.364	3.714	5.154	7.077

Table 7: Expected probabilities for each count of contracts over 10% of the cap

Conclusions

During the hypothesis testing, the expected probability of making the playoffs was slightly above 50% when a team spent an above average percentage of their total spending on forwards or defensemen, while the expected probability of making the playoffs was moderately below 50% when a team spent an above average percentage of their total spending on goaltenders. All three tests failed to reject the null hypothesis, however, meaning there is not significant evidence to suggest that spending an above average percentage on any of the three main positions yields a different probability of making the playoffs.

Similarly, the percentage of team cap spent on each position did not show any significance when predicting whether a team would make the playoffs. In fact, the only predictor besides the total cap was the amount spent on the top three forwards. This spending alone successfully predicted just over two thirds of season results. This implies that teams seeking to make the playoffs would do well to spend big on several high end forwards. Beyond that, however, the salary distribution among the other positions and lines does not appear to have as great of an effect, which means that teams would not find any boon in factoring in how much they are already spending on depth positions.

The hypothesis testing for the number of contracts over 10% of the total salary cap showed that having zero or one such contract led to below average playoff chances, and having two or more contracts led to above average chances, with the case of zero and three or more being significantly so. It is (as always) important to remember that correlation does not inherently imply causation, particularly in this case where the number of contracts over 10% of the cap is positively correlated with total money spent, which has already been shown to have a positive relationship with playoff probability. Even still, it appears that having extra contracts over 10% of the cap does not negatively affect a team's playoff chances. If a team has an opportunity to sign a very good player (who would require a large contract) who otherwise fits the team's needs, they should not avoid signing them even if they already have a number of large contracts on the books.

Using this metric to predict the number of playoff wins a team got also led to interesting results. Teams who had more players making over 10% of the cap were generally expected to win more playoff games, with the exception of teams with one such player having a slightly lower expectation than teams with no such players. This slight anomaly can largely just be attributed to the small sample size, since there weren't many teams with one or fewer such players. What was interesting, though, was the significant jump in the expected number of playoff wins for teams who had three or more highly paid players. What makes this jump interesting is the fact that the total salary was not significant for predicting playoff wins. This implies that having a lot of highly paid players could very well contribute to a team's expected playoff success. With this in mind, teams are definitely recommended to try and acquire a larger number of elite, highly paid players. There does not appear to be much benefit to keeping balance throughout the roster.

Trying to model the number of playoff wins, however, did not provide anything close to significant. All of the models were bad, and I would not choose any of them to try and predict playoff advancement. Total money spent, positional spending, or line by line spending do not have any significant effect on a team's playoff advancement. Teams should focus their roster building in these ways only to maximize their chances of making the playoffs, as playoff advancement appears to be determined by other factors.

Lastly, playoff teams who did not pay their backup goaltenders above the median amount appeared to find a bit more playoff success than teams who did pay their backups more. The question of whether you should seek a strong backup goaltender appears to be more a question of how much risk a team is willing to take in the playoffs. A lot of teams go through the entire postseason only ever starting their primary goaltender, in which case it makes sense that there is no benefit to paying your backup more. However, if a team's starting goaltender gets injured, they would be much better off if they had a better backup goaltender. This factor could serve to explain why teams with a well paid backup saw results that tended towards the middle when compared to those who did not, as having a stronger backup gives you a bit more of a safety net, but it comes at the cost of some money that you could be spending elsewhere on the roster.

In conclusion, NHL teams are heavily incentivized to make and perform well in the playoffs, both for pride and for money. According to the data, a team's chances of making the playoffs and then performing well in said playoffs are not affected by the percentage of their spending allocated to each position. Teams who spend more on their team as a whole and particularly those who acquire a strong top line of forwards stand an above average chance of making the postseason. This spending, however, becomes much less impactful once the playoffs begin. During the playoffs, having three or more players that are paid at least 10% of the cap was the only factor found to strongly increase the number of playoff games the team will win.

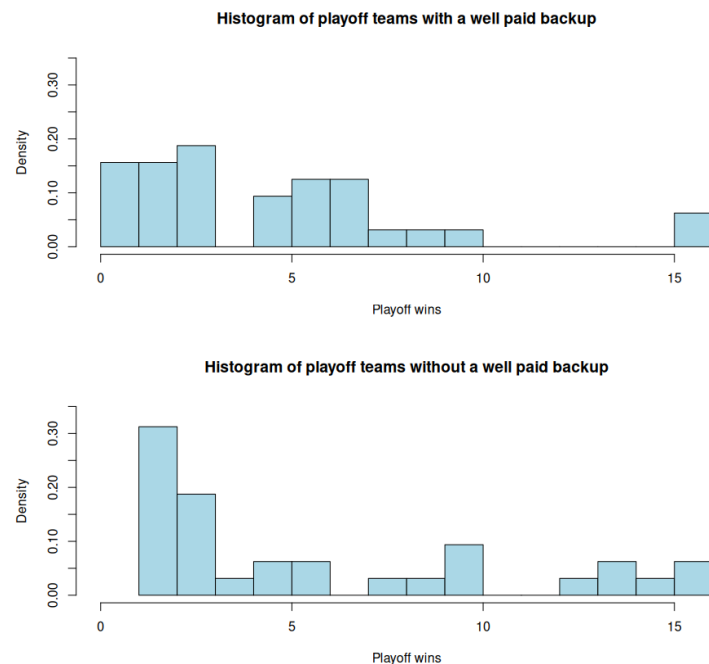


Figure 5: Comparison of playoff results between teams with or without a well paid backup goaltender

Appendix A: Unused Figures

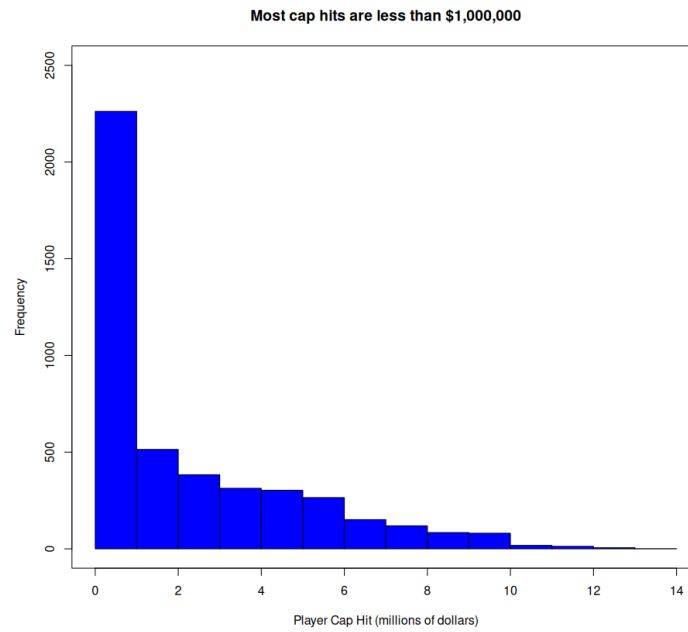


Figure A.1: Histogram of player contracts

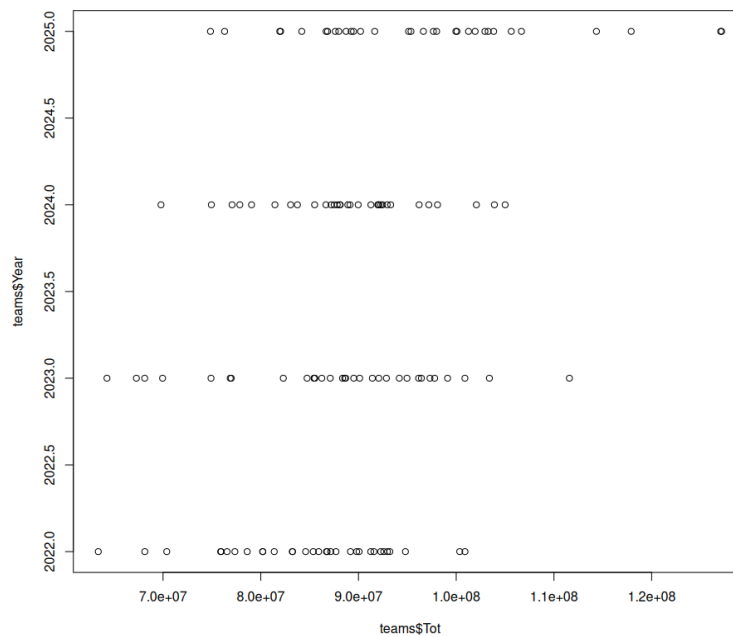
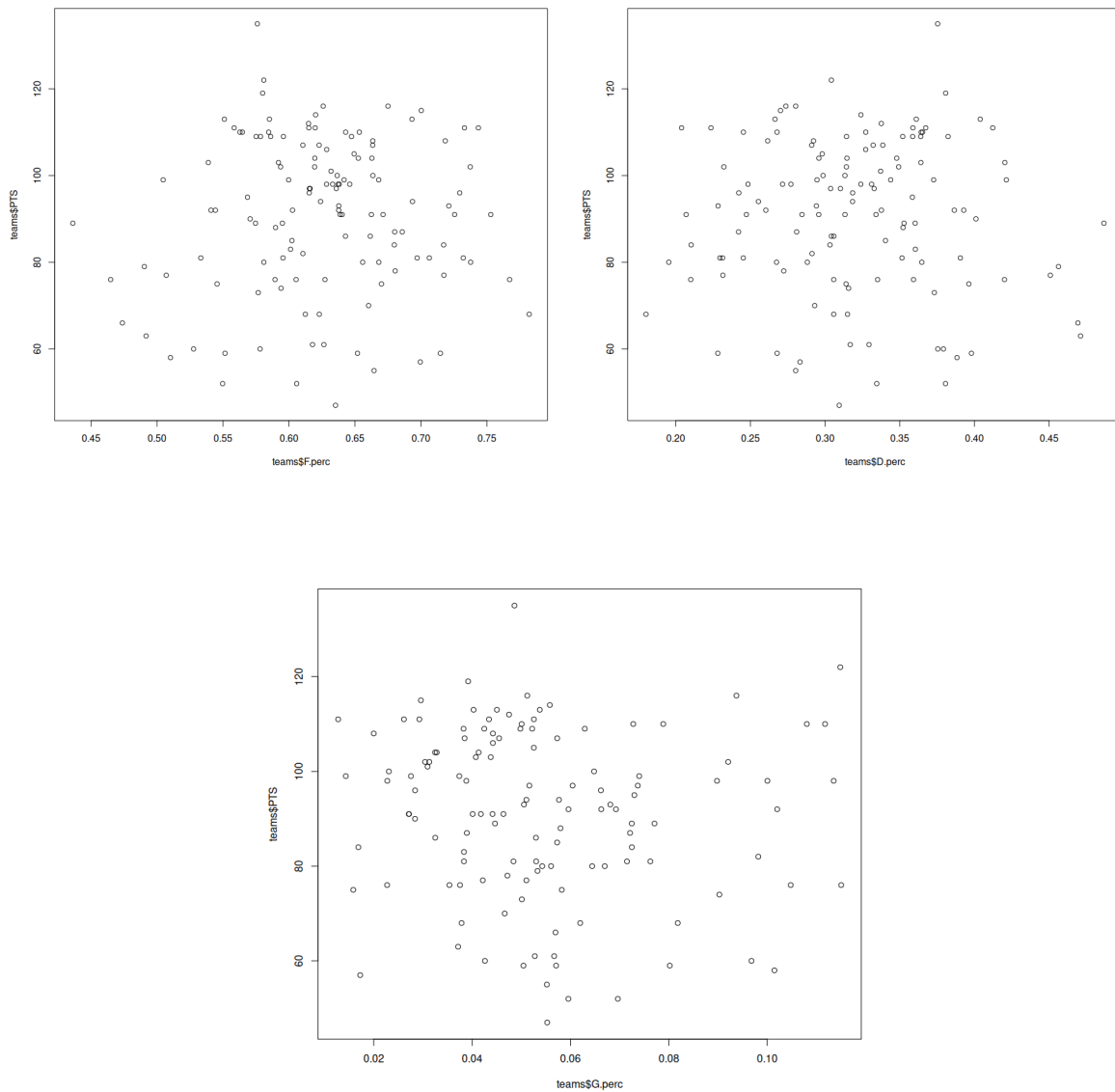


Figure A.2: Plot of total spending by year



Figures A.3a,b,and c: Plots of the relationship between season points and the percentage of a team's cap spent on forwards, defensemen, and goaltenders, respectively

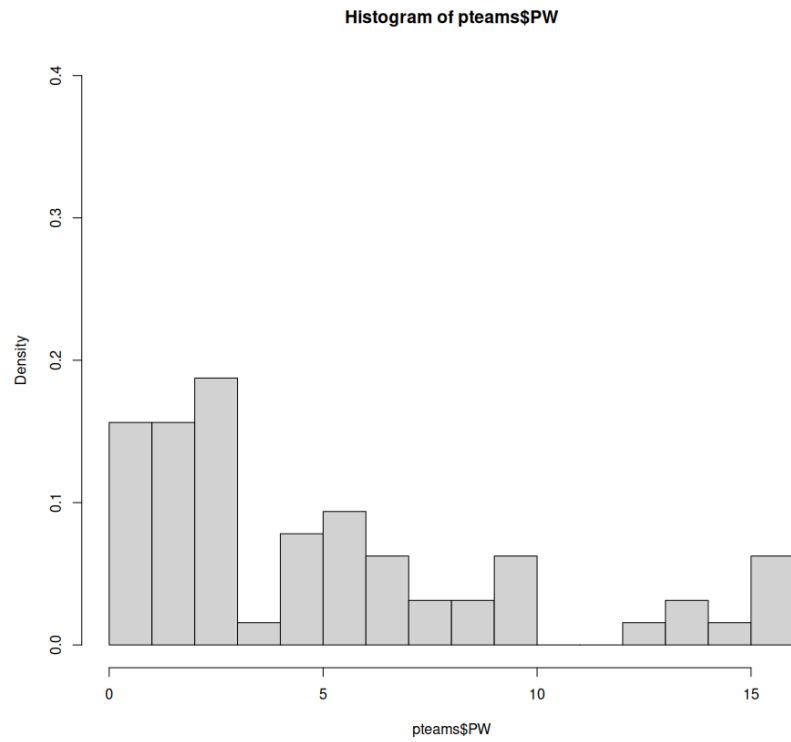


Figure A.4: Histogram of the playoff wins for all playoff teams

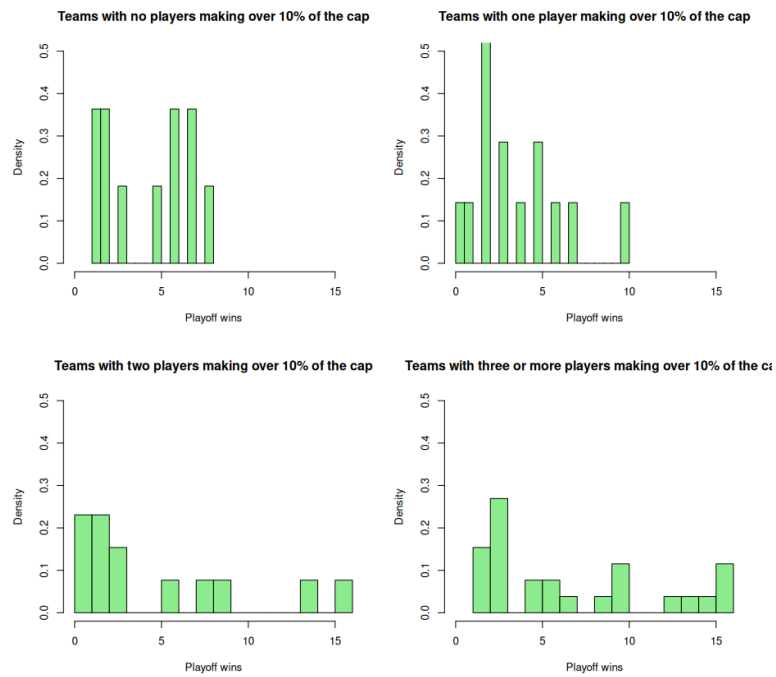


Figure A.5: Histograms of playoff wins for teams with different numbers of highly paid players

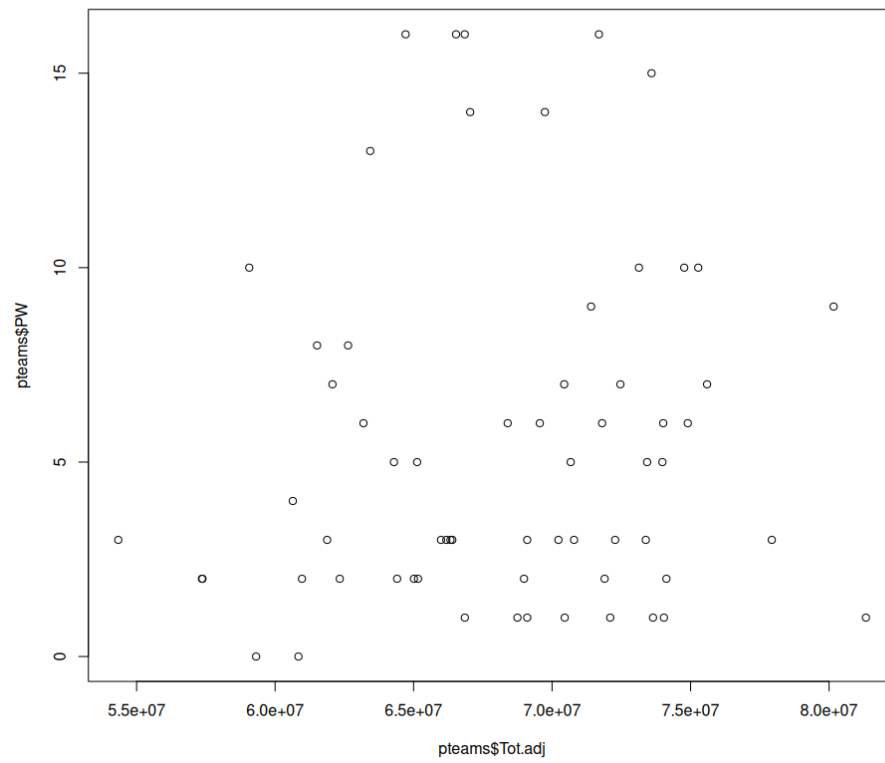


Figure A.6: Plot of the relationship between playoff wins and total adjusted spending

Appendix B: Background Information

NHL Team Structure:

NHL teams have a game day roster of twenty players: twelve forwards, six defensemen, and two goaltenders. The forwards are split into four lines of three players each, and the defensemen are split into three pairings of two. One of the goaltenders will typically be designated as the team's starting goaltender, and the other will be designated as their backup goaltender.

NHL League and Playoff Structure:

There are 32 teams in the NHL, split evenly into two conferences with two divisions in each conference. The top three teams in each division will make the playoffs, as will the top two teams in each conference that are not the top three in either division. This makes for a total of sixteen teams in the playoffs, exactly half of the total teams.

In the playoffs, teams will face off against each other in a sequence of best of four series. The winning team of each series will move on to the next round where they will play winners of another series, and the losing team will be eliminated. This continues for a total of four rounds, at the end of which there will be only one team remaining: the Stanley Cup Champion.

NHL Salary Cap and Floor:

Both the salary cap and floor increase every season. Below is a table of those values for the four seasons included in this project.

Season	Salary Floor	Salary Cap
2021-22	\$60.0 million	\$81.5 million
2022-23	\$61.0 million	\$82.5 million
2023-24	\$61.7 million	\$83.5 million
2024-25	\$65.0 million	\$88.0 million

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